College of Information Technology
CCM101 – Cloud Computing
CLOUD COMPUTING LABORATORY ACTIVITIES

Academic Integrity Policy
To uphold honesty, fairness, and academic excellence, all students are expected to follow these guidelines:

- **Cite your sources properly.** Any information, ideas, images, data, or words that are not your own and are not considered common knowledge must be properly cited.
- **Submit original work.** Your assignments should reflect your own understanding, analysis, and effort. Copying from books, websites, classmates, or any other source without proper acknowledgment is considered plagiarism.
- **Use AI responsibly.** AI tools (e.g., ChatGPT) may be used to brainstorm ideas, improve grammar, clarify concepts, or support learning. However, they should assist your thinking—not replace it.
- **Maintain meaningful authorship.** You are responsible for all submitted work. Assignments generated primarily by AI without your own analysis, revision, and proper disclosure constitute academic dishonesty.
- **Be transparent.** If AI or other external tools significantly contributed to your work, disclose their use when required by the instructor.
- **Ensure academic accountability.** Submission records, including timestamps and revision history where applicable, may be reviewed to verify compliance with this policy.

Guiding Principle: *Be the pilot of AI, not the passenger. Use technology to enhance your learning, but let your own knowledge, judgment, and critical thinking guide your work.*

Laboratory Activity 2

Mission 2: Build the Cloud Infrastructure Blueprint

Mission Overview
Congratulations,
Your onboarding has been successfully completed, and your Cloud Computing Portfolio has been approved by your supervisor.
CloudNova Technologies has now assigned you to your **first official project**.
Before deploying cloud services, every cloud engineer must understand the infrastructure that powers modern cloud computing. Your mission is to investigate the components of cloud infrastructure, identify how compute, storage, networking, and identity services work together, and document your findings as if you were preparing technical documentation for a client.
Using the **KillerCoda Playground**, Linux tools, official cloud documentation, and your GitHub Cloud Computing Portfolio, you will complete a series of engineering tasks that simulate the planning phase of a cloud deployment.
Remember: **Great cloud engineers build systems—but exceptional cloud engineers document and justify every design decision.**
.

Mission Objectives
At the end of this laboratory activity, you should be able to:

- Explain the major components of cloud infrastructure.
- Investigate the hardware and software resources available in a Linux environment.
- Differentiate compute, storage, networking, and identity resources.
- Interpret the relationship between cloud infrastructure components.
- Create professional technical documentation using Markdown.
- Continue building a structured GitHub Cloud Computing Portfolio.

Required Resources

- KillerCoda Playground
- GitHub Account
- Modern Web Browser
- Stable Internet Connection

College of Information Technology
CCM101 – Cloud Computing
CLOUD COMPUTING LABORATORY ACTIVITIES

Mission Rules

1. Complete each checkpoint in sequence.
2. Perform all technical tasks inside the KillerCoda Playground unless instructed otherwise.
3. Record your findings using Markdown.
4. Take screenshots as evidence for each completed checkpoint.
5. Commit and push your work to GitHub after completing each checkpoint.
6. Maintain professionalism in your documentation.
7. Academic honesty is expected. Collaboration is encouraged, but submitted work must be your own.

Mission Brief

A small company is planning to migrate its services to the cloud.
Your supervisor has asked you to prepare a **Cloud Infrastructure Assessment Report** before any servers are deployed.
You have been granted temporary access to a Linux server running in the cloud (KillerCoda). Your responsibility is to inspect the environment, identify infrastructure components, and prepare documentation that will help senior engineers design the final cloud architecture.

Mission Tasks

Checkpoint 1 – Expand Your Cloud Portfolio

1. Inside your existing GitHub repository, create a new folder named:

Laboratory-02-Build-the-Cloud-Infrastructure-Blueprint

2. Within this folder, create the following structure:

```
Laboratory-02-Build-the-Cloud-Infrastructure-Blueprint
│
├── README.md
├── infrastructure-report.md
├── cloud-components.md
├── cloud-provider-comparison.md
├── reflection.md
└── screenshots
```

3. Commit and push your changes to GitHub.

Checkpoint 2 – Investigate the Cloud Server

Using the KillerCoda terminal, investigate the Linux environment.
Determine and document:

- Operating System
- Kernel Version
- CPU Model
- Number of CPU Cores
- Total RAM
- Disk Capacity
- Mounted File Systems
- Hostname
- IP Address

Record your findings in:
infrastructure-report.md
Capture screenshots of your investigation.

Checkpoint 3 – Identify Cloud Infrastructure Components

Using your observations and the concepts discussed in **Chapter 2**, identify examples of the following infrastructure components found in your Linux environment:

- Compute Resources
- Storage Resources
- Networking Resources
- Operating System

For each component:

- Describe its purpose.
- Explain why it is important in cloud computing.
- Relate it to the Linux environment provided by KillerCoda.

Document your answers in:
cloud-components.md

Checkpoint 4 – Research the Major Cloud Providers

A cloud engineer must understand how different cloud providers offer similar infrastructure services under different names. Your task is to research the official documentation of the three leading public cloud providers and compare their core infrastructure services.

Visit the official documentation of the following cloud providers:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

Inside your laboratory folder, create a Markdown file named:
cloud-provider-comparison.md

Using your own understanding and the official documentation, create a comparison table showing the equivalent cloud services offered by each provider.

Your table must include the following infrastructure components:

Infrastructure Component	AWS	Microsoft Azure	Google Cloud Platform
Compute			
Storage			
Networking			
Identity and Access Management (IAM)			

After the table, answer the following guide questions in **2–3 sentences each**:

1. Which cloud provider offers the broadest range of services? Explain your answer.
2. Which cloud platform would you recommend for an organization that primarily uses Microsoft products? Why?
3. Which platform is widely recognized for Artificial Intelligence (AI), Machine Learning (ML), and Kubernetes services?
4. What similarities did you observe among the three cloud providers?

Use proper Markdown formatting with appropriate headings and tables.

Checkpoint 5 – Design a Simple Cloud Infrastructure

Create a simple cloud infrastructure diagram for a fictional company.

Your diagram should include:

- One Compute Resource
- One Storage Resource
- One Network
- One User
- One Internet Connection

You may use:

- Draw.io

- Excalidraw
- Figma
- Microsoft PowerPoint
- Any diagramming tool of your choice

Export the diagram as PNG.
Save it inside the folder:
screenshots
Name the file:
cloud-architecture.png

Checkpoint 6 – Technical Documentation
Update the laboratory README file.
Include the following sections:

- Mission Overview
- Objectives
- Cloud Infrastructure Components
- Tools Used
- Linux Commands Executed
- Skills Learned
- Challenges Encountered

Ensure the document follows proper Markdown formatting.

Checkpoint 7 – Mission Reflection
Create:
reflection.md
Answer the following questions:

1. Which cloud infrastructure component do you think is the most important? Why?
2. How does Linux support cloud computing?
3. Why is technical documentation important before deploying infrastructure?
4. What new skills did you learn during this laboratory activity?
5. How has your GitHub portfolio improved after completing this mission?

Your reflection should contain approximately **250–350 words**.

Expected Repository Structure

```
CCM101-cbadongen
├── README.md
├── Laboratory-01-Welcome-to-the-Cloud
└── Laboratory-02-Build-the-Cloud-Infrastructure-Blueprint
    ├── README.md
    ├── infrastructure-report.md
    ├── cloud-components.md
    ├── cloud-provider-comparison.md
    ├── reflection.md
    └── screenshots
        ├── server-information.png
        ├── network-information.png
        ├── storage-information.png
        └── cloud-architecture.png
```

Submission Guidelines
Submit the following:

- GitHub Repository URL

No ZIP files or email submissions will be accepted unless instructed otherwise. All deliverables must be available in your GitHub repository.

Mission Success Criteria
You have successfully completed Mission 2 if you can:

- Investigate a Linux server running in a cloud environment.
- Identify and explain the major components of cloud infrastructure.
- Compare equivalent cloud services offered by AWS, Microsoft Azure, and Google Cloud Platform.
- Create a simple cloud infrastructure diagram.
- Produce organized technical documentation using Markdown.
- Maintain a professional GitHub Cloud Computing Portfolio through meaningful commits and structured documentation.